

Tiny Bodies, Big Brains — Module 04: Cognition — Think About It

Part A: Family Discussion Questions

1. Where do thoughts come from? (Neurons firing in your brain!).
2. How many thoughts do you think you have in one day? (Thousands!).
3. What is "Object Permanence"? (Knowing something still exists even when you cannot see it). When did you learn this?
4. Why can you solve harder puzzles now than when you were younger? (Your brain has built more connections!).
5. What is the difference between remembering something and understanding it?
6. How does sorting things into groups help your brain? (It makes the world easier to predict).
7. What is the smartest thing your brain did today without you even trying? (Kept your heart beating, balanced your body, understood words).
8. Why is it harder to think when you are tired? (The brain needs energy and rest to work well).
9. Can you think about something you have never seen? (Yes! That is imagination — your brain creates new models).
10. What is your earliest memory? How old were you? Why do you think you remember that one?

Part B: Hands-On Activities and Reflection

1. Play the "Memory Tray" game: Put ten objects on a tray. Look for 30 seconds. Cover the tray. How many can you remember? Your working memory is being tested!
2. Sort a pile of buttons, toys, or snacks into groups. First by color, then by size, then by shape. Each time you sort, your brain is organizing its model of the world.
3. Play "Hide and Seek" with an object. Have someone hide it while you watch, then count to 20 and find it. Babies younger than 8 months cannot do this — their brain has not built object permanence yet!

4. Draw the same thing (a house, a tree, a person) at different ages. Compare your current drawing to one from years ago. Your brain's model got more detailed!
5. Try a new puzzle or brain teaser together. If you get stuck, talk through your thinking out loud. How does sharing thoughts help solve problems?
6. Close your eyes and imagine your favorite place. What do you see? Smell? Hear? Your brain can run a simulation without any real sensory input!
7. Count backward from 20 while hopping on one foot. Is it hard to do two things at once? Your brain has limited working memory — it is juggling tasks.
8. Play "Categories": Name five animals, five foods, five colors — as fast as you can. Your brain organizes knowledge into categories so it can find information quickly.
9. Ask a parent about something they believed as a child that turned out to be wrong (a misconception). What new data helped them update their model?
10. Before sleep, think of three things you figured out today. Your brain built new pathways while you were thinking. Sleep will make them even stronger!